Lab 07 Tasks

Q1: Image Transformations

(a) Apply **2D transformations** (translation, rotation, scaling, and shearing) on any image and observe how its position and shape change.

(b) Perform a **3D transformation** to visualize how an object or shape changes when rotated or moved along the x, y, and z axes.

(c) Represent a 2D point using **homogeneous coordinates** and explain why the extra coordinate is used in computer vision.

(d) Apply an **affine transformation** on an image and describe whether straight lines remain straight after transformation.

(e) Use **projective transformation (homography)** to correct the perspective of a tilted image, such as making a slanted book look flat.

Q2. Perspective Correction (Unwarping)

This is the classic computer vision problem of taking a photo of a flat surface (like a document, screen, or painting) taken at an angle and correcting it to appear as if it was photographed head-on.

Modify the apply_perspective_transform function's inputs to perform an **inverse** transformation: you'll take the tilted view as the source and map it back to a perfect rectangle as the destination.

Q3. Undistort Live Camera Feed

- Calibrate once, save the camera matrix and distortion coefficients.
- Use cv2.VideoCapture() to open your webcam.
- Apply real-time **undistortion** using cv2.undistort().

Q4. Build a 3D Model from Images Using COLMAP

Use **COLMAP** to automatically:

- Detect features
- Match them across images
- Estimate camera poses

- Construct a **3D scene** (point cloud + mesh)